

Artificial Intelligence (CS F407)

Credits: 3

Lecturer: Tirtharaj Dash (Room CC-107) **and** Ashwin Srinivasan (Room D-167)

Office-Hours: By appointment (send an email to fix a time)

Lecture and Lab Details

Day	Time	Location	What?
Tue	11-12	LT-2	Lecture
Wed	8-9?	LT-2?	Lect/Tutl
Fri	11-12	LT-2	Lecturel
Take Home			Proj etc.

Introduction

Pre-requisites (courses a student should have taken before this):

- Preferable: Data Structures and Algorithms

Post Condition (on student capability after successfully completing the course):

- At the end of the course, the student will be able to:
 1. Understand the basic concepts underlying Artificial Intelligence (AI) tasks, namely problem solving, search, representation, reasoning, decision making and learning;
 2. Able to understand use AI tools (like LLMs) to address old AI problems.

Lectures and Labs

In the tabulation below the number in brackets refers to the number of lectures for that topic.

Week	AI SCIENCE	AI ENGINEERING	Labs (NewFAI meets GOFAI)
1	Introduction (1)	Neural models (2)	Using LLMs to generate code for solving: • Brute-force search • Heuristic search • Constraint satisfaction
2	Agents and Search(2)	LLMs (1)	
3	Logical Intelligence (2)	Transformers (1)	
4	Planning (1)	LLMs-Agents (2)	QA with LLM-Agents
5	Probalistic Intelligence (1)	Extending LLMs: Tools(2)	
6	Probabilistic Intelligence(2)	Extending LLMs: RAG(1)	
7	Probabilistic Intelligence(2)	Fine-tuning(1)	
8	Neural Intelligence(2)	Beyond Transformers (1)	
9	Reward-based Intelligence(2)	RLHF(1)	Project
10	Learning (2)	Deep Learning (1)	
11	Guest lectures on Applications of AI (3)		
12	Project Presentations		
13	Project Presentations		

Assessment

	Assessment Component	Marks
1	Worksheets/Labwork (best k-of-N)	20
2	Comprehensive	30
3	Project	30
4	Paper Reading	10
5	Tutorial Participation	10

Notes:

- Lectures and tutorials will be conducted according to the schedule, with possible alterations announced in class. Each week you are usually expected to complete a worksheet with questions on the material for that week. These form part of the Assessment as shown above. NOTE: There is no mid-semester exam.
- The worksheets in will either be multiple-choice questions based on the material you are expected to know upto that week or hands-on laboratory work. The lab-work may be assessed by questions in the Comprehensive exam. This will be in the form of questions covering the topics covered in the hands-on assignments. Difficulty with a previous worksheet can be discussed in the Tutorial in the following week.
- The paper-reading component will be assessed by questions on a classic paper. The paper will be given to you in Week 1. There will be questions on the paper as an extra component of your Comprehensive exam.
 - We will read this paper: Turing, A. M. (2007). Computing machinery and intelligence. In *Parsing the Turing test: Philosophical and methodological issues in the quest for the thinking computer* (pp. 23-65). Dordrecht: Springer Netherlands. <https://academic.oup.com/mind/article/LIX/236/433/986238>
- Topics for projects will be made available by Week 2. Project work is to be done in groups (no more than 3). Project assessment details will be announced in class.

Bonus Marks Scheme

In addition to the above, there is a “bonus marks” scheme, that you can subscribe to, by reading a list of AI-related books. Bonus marks will be awarded based on whether you can convince the lecturers that you have indeed read the books. For this semester the books and their *maximum* bonus marks are shown in brackets:

- Non-fiction. Any or all of:
 - a. Yuval Noah Harari, *Nexus*. (3 marks)
 - b. Judea Pearl, *The Book of Why*. (3 marks)
 - c. Arvind Narayanan and Sayash Kapoor, *AI Snake Oil*. (2 marks)
- Fiction. At most 2 of:
 - a. Isaac Asimov, *Bicentennial Man* (1 mark)
 - b. Arthur C Clarke, *2001: A Space Odyssey* (1 mark)
 - c. Philip K. Dick, *Do Androids Dream of Electric Sheep* (1 mark)
 - d. Peter Watts, *Blindsight* (1 mark)
 - e. D.F. Jones, *Colossus* (1 mark)

Note: If the lecturers decide that you have not actually read the book but have relied on a summary from other sources (human or machine), you will be penalized the book’s maximum mark. That is, you will get (-X) marks for a book worth at most X marks if you are deemed to have attempted this deception.

Text

- **S. Russell and P. Norvig**, *Artificial Intelligence: A Modern Approach*, Prentice-Hall, 4th ed. (2021)
 - The most comprehensive introductory text around for AI

Other Useful Books

- Chip Huyen, *AI Engineering*, (2024), O’Reilly Associates.
- Sebastian Raschka (2024) *Build an LLM from Scratch*. Manning.
- R.E. Neapolitan and X. Jiang, *Artificial Intelligence with an Introduction to Machine Learning*, 2nd ed., (2018), Chapman and Hall.